RAJALAKSHMI ENGINEERING COLLEGE

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J2

Started on	Monday, 24 August 2026, 4:23 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 11:56 AM
Time taken	19 hours 33 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int count(int a[], int l, int r, int x)
{
    int c = 0;

    for (int i = l; i <= r; i++)
        if (a[i] == x)
            c++;

    return c;
}

int maj(int a[], int l, int r)
{
    if (l == r)
        return a[l];

    int m = (l + r) / 2;

    int x = maj(a, l, m);
    int y = maj(a, m + 1, r);

    if (x == y)
        return x;

    int cx = count(a, l, r, x);
    int cy = count(a, l, r, y);

    if (cx > cy)
        return x;
    else
        return y;
}
```

```
35
36  int main()
37  {
38      int n;
39
40      scanf("%d", &n);
41
42      int a[n];
43
44      for (int i = 0; i < n; i++)
45          scanf("%d", &a[i]);
46
47      printf("%d", maj(a, 0, n - 1));
48
49      return 0;
50  }
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course